

A Decodable Signalling Code Co-evolves but Stays Private to Kin: Survival Selection in a Minimal World

Chunxuan Yang

Sogang University, Seoul, Republic of Korea
yang2023@sogang.ac.kr yangchunxuan1@gmail.com

Abstract

Can a decodable signalling code *grow* from random weights under survival selection alone, with no gradient on the task, no oracle labels, and no borrowed weights, and is what grows public or private to kin? In a minimal hand-coded 2-D lethal-vent world, tiny tied code-table agents evolve a decodable channel. Decomposing the result against an architecture-matched random-fitness control (whose only difference is the reproduction-fitness signal), we find selection *sharpens* rather than originates the channel: clonal descent alone reaches mutual intelligibility (MII) 0.1174 ($\sim 73\times$ the $1/625$ chance floor), and survival selection adds a paired win-margin of $+0.0548$ (95% CI $[0.0234, 0.0849]$, $n=48$) at the pre-registered gen-28 fixation point. A post-hoc probe shows this margin is a *transient acceleration* that decays as the control's own clonal fixation catches up. The code stays private to kin (dominant-lineage share 99.1%, N_{eff} 1.02); even when diversity is actively sustained, and even when survival is made to depend on decoding non-kin, no public cross-lineage code bootstraps, a *pressure-vs-survival tension*. We also contribute a bidirectional self-audit that catches deflation as readily as inflation.

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Code & verdict artifacts: <https://github.com/yangchunxuan/grown-grounded-language>; every result regenerates from committed, seeded verdict JSONs.

The question and the world

Where does a shared signalling code come from? Most computational accounts *train* a channel in (gradient descent on a task with a differentiable channel; Foerster et al., 2016; Lazaridou et al., 2017) or *design* one in (an imposed iterated-learning transmission bottleneck; Kirby et al., 2008; Ren et al., 2020). We take up the prior question: can a decodable code *grow* from random weights under **survival selection alone**, and if it does, is it a **public** code or a code private to **kin**?

The substrate is deliberately minimal and entirely hand-coded: a 48×48 lethal-vent world in which agents metabolise, can die, and reproduce, and in which a single tied UnifiedIO code-table ($4 \times 5 \times 6$, $\mathcal{N}(0,1)$ init) is the only communicative apparatus. There is no gradient on communication, no oracle (agents never read world truth), and no borrowed weights. We measure mutual intelligibility (MII) behaviourally: route a referent through one agent's emit into another's decode and score exact-tuple recovery against a chance floor of $1/625 \approx 0.0016$. Within-founder (WF) and cross-founder (CF) MII separate a *private kin code* (WF high, CF at chance) from a *public* one (CF also above chance).

Co-evolution is real, but decomposes and stays kin-private

The control is not communication-ablated: both arms carry the full code-table and run the identical loop, and the only manipulated variable is the reproduction-fitness signal (survival vs. uniform-random, with the reproduction RNG byte-matched). This isolates selection from *clonal descent-convergence*. The random-fitness arm also fixates to $\sim$ one lineage, so its MII of **0.1174** ($\sim 73\times$ floor) is the descent baseline, not zero. At gen 28 the survival arm reaches **0.17219**, a paired win-margin of $+0.0548$, **95% CI** $[0.0234, 0.0849]$ ($n=48$; Figure 1). Selection therefore *sharpens* a largely descent-determined channel rather than originating it.

A post-hoc transient probe ($n=24$; gen 28/56/100/150) shows this margin is a *transient acceleration*: the survival-arm MII stays ~ 0.16 – 0.17 while the control's MII rises ($0.117 \rightarrow 0.138$) as clonal fixation accumulates cross-agent decodability. The paired margin decays from $+0.0569$ at gen 28 (CI excludes zero) to $+0.0249$ at gen 150, with the CI including zero from gen 100 onward. The headline is therefore scoped to 28 generations.

The code is *not public*. The survival arm fixates to a single founder lineage (dominant-lineage share **99.1%**, N_{eff} **1.02**, 47/48 seeds single-lineage), so WF runs ≈ 0.16 ($\sim 100\times$ chance) while CF sits at the floor.

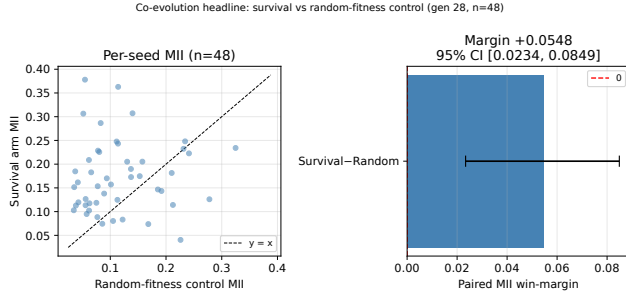


Figure 1: Co-evolution headline ($n=48$, gen 28): per-seed survival vs. random-fitness MII (left) and the paired win-margin $+0.0548$, 95% CI $[0.0234, 0.0849]$, which excludes zero (right). Source: committed verdict JSON.

The wall: no public code bootstraps

Is kin-privacy merely an artifact of fast founder collapse? We sustained genuine multi-lineage coexistence by niching (tournament selection plus lineage fitness-sharing) at $\text{pop}=96$: the soft96 cell holds N_{eff} 4.10 over 20+ coexisting generations, yet CF stays at the floor (CF 0.00153 vs. floor 0.0016; the CF–floor CI straddles zero). Maintaining diversity is *necessary but not sufficient*.

Making survival itself depend on decoding non-kin does not help either. Across three heterogeneous interventions (Table 1), CF never rose above chance; where the pressure bit, it crashed survival. Non-kin codes are mutually undecodable, so forced cross-listening is acting on noise, which starves foraging: a *pressure-vs-survival tension*. All cross-lineage arms are $n=8$ (positive-or-inconclusive tier), so we report that a public code *did not* bootstrap under the regimes tested, never that none *can*.

Table 1: Cross-lineage interventions ($n=8$ each): CF stays at or below the chance floor (≈ 0.0016); where pressure bites, survival crashes.

Intervention	frac. alive	CF
C2X2 (0.25 non-kin, viable)	0.157	0.00156
C2X3 (0.5 non-kin, quota)	0.100	0.00192
C2X (100% non-kin, cold)	0.046	0.00166

A bidirectional self-audit

Each claim must survive an architecture-matched control, a fair baseline, ≥ 6 -seed bootstrap-CI win-margins, frozen pre-registration, a no-oracle redline, and code-grounded adversarial review. The audit’s single methodological claim is *bidirectionality*: it is built to catch deflation as readily as inflation. Two in-scope catches illustrate this. (1) A pilot/formal false *negative*: a formal treatment arm had been compared against a pilot control, burying a real effect;

the load-bearing knob was generation count. (2) An RNG-offset bug in the headline control, whose fix left the conclusion intact and slightly *strengthened* the margin ($+0.045 \rightarrow +0.0548$). Catching a deflation is the direction conventional self-checking rarely covers.

Contribution and scope

A decodable code grows from scratch under survival selection but stays kin-private; selection *accelerates* a clonal-descent channel rather than originating a public one; and no public code bootstraps even under sustained diversity and direct cross-lineage pressure. The results are scoped to one hand-coded world, one architecture, and per-stage timescales (28–48 generations). We read the wall as motivating a richer world and a consequence-driven learner rather than more selection pressure. All code and seeded verdict JSONs are committed and reproducible.

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